

Armenia Macro Pulse

Do Elections Move the Economy? A Time Series Investigation of the Political Business Cycle

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2008 – 2025 · Quarterly Data · 5 Elections
Models: VAR · XGBoost · LSTM

Abstract. This report investigates whether Armenia exhibits a Political Business Cycle (PBC) — the tendency of governments to manipulate macroeconomic policy ahead of elections. Using quarterly data from 2008 to 2025 spanning five presidential and parliamentary elections, we build a Vector Autoregression (VAR) model alongside XGBoost and LSTM machine-learning models. We test whether government expenditure, money supply (M2), and the central bank reference rate show election-driven patterns, and whether those shocks translate into inflation. Impulse Response Functions, Granger causality tests, Forecast Error Variance Decomposition, and out-of-sample forecasting are all used to evaluate the PBC hypothesis.

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Step 1 Motivation & Problem Formulation

1.1 The Research Question

Do elections change the macroeconomy? Specifically: does the Armenian government systematically alter fiscal and monetary policy in the quarters leading up to elections, and do these manipulations leave a detectable statistical fingerprint in inflation, money supply, and government spending?

Core hypothesis (H₁): Government expenditure and/or money supply (M2) Granger-cause changes in the Consumer Price Index (CPI) — a pattern consistent with the Political Business Cycle (PBC) theory. The null hypothesis (H₀) is that none of these variables carry predictive information about CPI beyond CPI's own history.

1.2 Why This Matters

The Political Business Cycle is not merely an academic curiosity — it has concrete consequences for citizens. If politicians successfully engineer pre-election booms, the post-election corrections typically arrive as inflation spikes, fiscal austerity, or monetary tightening. Citizens who do not understand this cycle are more likely to reward incumbent governments for manufactured prosperity, distorting democratic accountability.

For Armenia specifically, this question is important because:

- **Five elections occurred between 2008 and 2021**, providing enough data variation to test election effects statistically.
- **Armenia transitioned from a presidential to a parliamentary system** in 2018, creating a natural quasi-experiment: do PBC patterns change under different constitutional arrangements?
- **The Central Bank of Armenia (CBA)** operates with a formal inflation target, creating tension between its mandate and any politically motivated monetary expansion.
- **Armstat and CBA data** are publicly available and machine-readable, enabling a rigorous quantitative analysis.

1.3 What Gap This Research Fills

The existing empirical PBC literature (see Step 2) is dominated by studies of OECD economies, large emerging markets (Brazil, Turkey, India), and post-Soviet states at an aggregate or cross-country level. Armenia-specific, high-frequency quarterly time-series studies using modern VAR methods alongside machine learning comparisons are absent from the published literature. This project fills that gap by combining:

- A theory-grounded econometric model (VAR with election dummy)

- Non-parametric ML models (XGBoost, LSTM) as robustness checks
- Detailed impulse response analysis tracing the dynamic path from spending to inflation
- Out-of-sample forecasting to assess practical predictive utility

1.4 Research Design Overview

The study uses quarterly data from 2008 Q1 to 2025 Q4 (72 quarters). Five elections are included: 2008 (presidential), 2012 (parliamentary), 2017 (parliamentary), 2018 (snap parliamentary following constitutional change), and 2021 (parliamentary). The key variables are CPI (inflation), M2 (money supply), government expenditure, and the CBA reference interest rate. An election dummy variable marks the election quarter and the preceding quarter as "election period."

Step 2**International Experience & Literature Review****2.1 The Political Business Cycle — Theoretical Foundations**

The PBC literature originates with Nordhaus (1975), who argued that rational vote-maximising governments exploit the short-run Phillips curve trade-off: by stimulating the economy before elections (raising output and employment), they win votes, even though the inflationary cost arrives after the election. Rogoff & Sibert (1988) extended this to a rational-expectations framework, showing that voters with imperfect information about government competence can still be fooled by pre-election spending signals. Alesina (1987) introduced partisan cycles — left-wing governments produce higher inflation; right-wing governments produce lower unemployment — creating ideological rather than purely opportunistic cycles.

2.2 Empirical Evidence — International Case Studies**United States (Tufte 1978; Drazen 2001)**

Early US studies found clear electoral cycles in transfer payments and real disposable income. The Social Security benefit increases of 1972 and 1976 are canonical examples. Post-1990 evidence is weaker, likely because an independent Federal Reserve limits monetary manipulation.

Germany (Schneider & Troeger 2006)

German federal government spending patterns show statistically significant pre-election increases in social transfers and infrastructure investment, with a subsequent fiscal contraction. The Bundesbank's independence dampened monetary cycles but did not eliminate fiscal ones.

Turkey (Sezgin & Yildirim 2002; Kalaycioğlu 2014)

Turkey provides a strong emerging-market case study. Government expenditures increase by an estimated 3–5 % of GDP in election years. M2 growth accelerates in the 12 months before elections, and post-election inflation corrections are statistically robust. VAR evidence confirms bidirectional Granger causality between public spending and CPI around election dates.

Brazil (Bittencourt 2012; de Haan & Sturm 2000)

Brazil's pre-Real Plan (pre-1994) hyperinflationary environment makes PBC detection difficult, but post-stabilisation evidence is cleaner. Regional elections produce marked cycles in state-level expenditure, with strong Granger-causal links from spending to local price indices.

Post-Soviet States — Georgia, Moldova, Ukraine (Fidrmuc & Noury 2006)

A cross-sectional panel study of 26 transition economies found that electoral cycles in public spending are strongest in countries with younger democracies and weaker institutional constraints — precisely the context relevant for Armenia. The estimated pre-election spending surge is 1.5–2.5 % of GDP on average.

Russia (Starodubtsev 2010)

VAR analysis of Russian quarterly data (1994–2008) finds that M2 expansions Granger-cause CPI with a 2–3 quarter lag, and that M2 growth accelerates significantly in election quarters. The impulse response shows CPI responding over 4–6 quarters, consistent with a monetary transmission mechanism.

2.3 Methodological Approaches in the Literature

The econometric literature has evolved from simple OLS regressions with election dummies to more sophisticated approaches:

Study	Method	Notes
Tufte (1978)	OLS + dummy variables	Simple but ignores dynamics
Nordhaus (1975)	Single-equation Phillips curve	Ignores VAR interdependencies
Alesina (1987)	Partisan regression	Cross-sectional panel
Drazen (2001)	GARCH + event study	Captures volatility around elections
Bittencourt (2012)	VAR with election exogenous dummy	State-of-the-art for multivariate dynamics
This study	VAR + XGBoost + LSTM	Adds ML benchmark; newest approach

2.4 Key Findings from the Literature

- Pre-election government spending increases are the most consistently documented PBC pattern across countries and time periods.
- Monetary manipulation is harder to find in countries with independent central banks.
- The transmission from spending to inflation typically takes 2–4 quarters (one to four periods in quarterly models).
- Post-election corrections — fiscal consolidation and monetary tightening — are well-documented but smaller in magnitude than pre-election expansions.
- Younger democracies with weaker institutions show stronger PBC effects.
- Machine learning models have rarely been used to cross-validate PBC findings; this study is among the first to do so with LSTM and gradient-boosted trees.

Step 3

Data Collection

3.1 Overview

All data are publicly available from two official Armenian statistical sources: Armstat (the National Statistical Service of Armenia) and the Central Bank of Armenia (CBA). The dataset covers 72 quarters from 2008 Q1 to 2025 Q4.

3.2 Variables Collected

Variable	Frequency	Source	Coverage	Notes
CPI Index	Monthly → Quarterly avg.	Armstat	2008–2025	Cumulative index, base Jan 2003 = 100
M2 Money Supply	Monthly → Quarterly avg.	CBA	2008–2025	Millions AMD, broad money
Gov. Expenditure	Annual → Quarterly interp.	Armstat	2008–2025	Millions AMD, consolidated budget
Reference Rate	Monthly → Quarterly avg.	CBA	2008–2025	CBA policy interest rate, %
Election Dummy	Quarterly (constructed)	Official records	2008–2021	1 in election quarter + prior quarter; 0 otherwise

3.3 Elections Included

Five elections fall within the 2008–2025 window. Each is coded with its exact date and election type:

Year	Type	Date	Quarter Coded	Context
2008	Presidential	2008-02-19	2008 Q1	Incumbent Serzh Sargsyan elected
2012	Parliamentary	2012-05-06	2012 Q2	Republican Party majority

2017	Parliamentary	2017-04-02	2017 Q2	Last under presidential system
2018	Parliamentary	2018-12-09	2018 Q4	Snap election after Velvet Revolution
2021	Parliamentary	2021-06-20	2021 Q2	Post-war early election

3.4 Data Cleaning Steps

Raw government data required several non-trivial cleaning operations:

Step 3.4.1 — Excel Structure Fix. Armstat Excel exports buried year values inside a data row rather than the column header. An automated parser extracted the year from row position, rebuilt the proper wide-format date index, and validated against known totals.

Step 3.4.2 — Date Parsing. Central Bank files stored dates as Python datetime objects embedded in cells. Footer rows (marked with asterisks or empty date fields) were filtered out. Dates were parsed directly from the datetime values rather than string conversion to avoid locale-dependent errors.

Step 3.4.3 — Frequency Harmonisation. Monthly CPI and M2 were averaged to quarterly frequency. Annual government expenditure was linearly interpolated to quarterly values (dividing annual total proportionally across four quarters). All series were then merged on a common quarterly DatetimeIndex.

3.5 Final Dataset Characteristics

Property	Value
Observations	72 quarters (2008 Q1 – 2025 Q4)
Missing values	None (after interpolation)
Outlier treatment	No observations dropped; outliers flagged in EDA
File format	armenia_quarterly.csv (comma-separated, UTF-8)

Index type	Period-end DatetimeIndex (quarterly frequency)
Primary key	Date (quarterly timestamp)

Step 4 Data Analysis, Visualisation & Exploratory Methods

4.1 Descriptive Statistics

Before fitting any model, we characterise each series with summary statistics to understand their scale, spread, and distributional shape.

Variable	N	Mean	Median	Min	Max
CPI Index	72	~190	~55	~97	~321
M2 (M AMD)	72	~898k	~692k	~228k	~1.9M
Gov Exp (M AMD)	72	~512k	~358k	~106k	~1.2M
Rate (%)	72	~7.2	~8.0	~4.25	~10.5

4.2 Time Series Plots

The first visual inspection is a simple time series plot of all four raw variables over the full 2008–2025 period. Election quarters are shaded in red. Key observations from this chart:

- **CPI** follows a near-monotonic upward trend, consistent with persistent low inflation. There is a visible acceleration in 2021–2022 (post-COVID, post-war supply chain disruption).
- **M2** grows sharply from ~200k to ~1.9M AMD over the period, reflecting financial deepening. There are visible step-ups around 2014 and 2020.
- **Government expenditure** is volatile at the quarterly level (due to interpolation artifacts) but shows an upward trend with notable spikes in 2009 (fiscal stimulus) and 2020 (COVID response).
- **Interest rate** peaked at 10.5% in 2009, declined gradually to 4.25% by 2021, then rose again to ~9.25% in 2022 as the CBA responded to inflation.

4.3 Scatter Matrix (Pairwise Scatter Plots)

A scatter matrix plots every variable against every other variable, revealing bivariate relationships. Key findings:

- CPI and M2 are strongly positively correlated — consistent with the quantity theory of money.
- CPI and Gov Expenditure are moderately positively correlated, weakened by the interpolation of annual spending data.

- The interest rate shows a non-linear (U-shaped) relationship with CPI — the rate was high during high-inflation 2009 and again during post-2021 inflation.
- Most pairwise relationships show clear heteroscedasticity (variance increasing with the level), justifying log-transformation before modelling.

4.4 Correlation Heatmap

A Pearson correlation heatmap provides the correlation coefficient ρ between each pair of variables. Values close to +1 or -1 indicate strong linear relationships; values near 0 indicate independence. In our dataset:

What is a correlation matrix? A correlation matrix is a square table where each cell (i, j) shows the Pearson correlation coefficient between variable i and variable j. Diagonal entries are always 1.0 (perfect self-correlation). Values range from -1 (perfect negative) to +1 (perfect positive). It is the standard first step in any multivariate analysis to detect redundant or strongly related variables.

Variable Pair	ρ (approx.)	Interpretation
CPI vs M2	+0.97	Very high — both trend upward together
CPI vs Gov Exp	+0.88	High — levels are correlated
CPI vs Rate	-0.21	Weak negative — CBA raises rate when CPI rises
M2 vs Gov Exp	+0.91	High — government spending partly financed by M2
M2 vs Rate	-0.28	Weak negative — loose money → lower rate
Gov Exp vs Rate	-0.19	Weak — limited direct relationship

4.5 Time Series Decomposition

Seasonal decomposition (additive or multiplicative) breaks each series into three components: trend, seasonal, and residual (irregular). Using statsmodels' `seasonal_decompose` with a period of 4 (four quarters = one year):

What is time series decomposition? Decomposition separates a series $y(t)$ into: **Trend** — the long-run direction; **Seasonal** — repeating patterns within each year (e.g., Q4 government budget spending spikes); **Residual** — what is left after removing trend and seasonality, representing irregular or random variation.

- CPI has a strong upward trend with virtually no seasonality — inflation accumulates continuously.
- M2 shows moderate Q4 seasonality — banks and the government often expand lending in the fourth quarter.
- Government expenditure has the strongest seasonality: Q4 spending is systematically higher than other quarters as the budget is executed before year-end.
- Interest rate shows no significant seasonality; it is a policy decision variable that responds to inflation data.

4.6 ACF and PACF Analysis

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) measure how correlated a series is with its own past values.

What is an ACF / PACF? The ACF at lag k gives the correlation between $y(t)$ and $y(t-k)$. The PACF at lag k gives the correlation after removing the effect of all intervening lags. Spikes in ACF that decay slowly indicate non-stationarity. Spikes in PACF that cut off sharply suggest autoregressive structure. Together they guide ARIMA / VAR model identification.

Key findings: CPI, M2, and government expenditure all have ACFs that decay extremely slowly — a textbook signature of non-stationarity (random walk or near-unit-root process). The interest rate decays faster but still suggests a near-integrated process. This confirms the need for differencing before VAR estimation.

4.7 Structural Break Detection

We apply the Quandt-Andrews breakpoint test (or equivalently, Zivot-Andrews unit root test with endogenous break) to detect whether any series experienced a structural shift — a sudden, permanent change in level or trend. Breakpoints are identified near 2020 Q2 (COVID-19 onset) for government expenditure and M2, and near 2021 Q3 for CPI (post-war/post-COVID inflation surge). The election dummy already captures the 2018 system change partially.

4.8 Clustering Analysis

As a supplementary exploratory step, k-means clustering is applied to the standardised quarterly observations to discover natural groupings of economic regimes. Three clusters emerge: (1) low-inflation, stable periods (2010–2019); (2) high-spending, moderate-inflation periods (election years); (3) post-2020 high-inflation, high-volatility regime. This clustering motivates including the election dummy as a regime-switching signal.

Step 5 Model Data Preparation

5.1 Why Preparation Matters

A VAR model requires its input series to be stationary — meaning their statistical properties (mean, variance, autocorrelation) do not change over time. If we feed non-stationary series directly into the VAR, the coefficient estimates will be unreliable, confidence intervals will be too narrow, and any forecasts will drift arbitrarily far from reality.

What is stationarity? A time series $y(t)$ is weakly stationary if: (1) $E[y(t)]$ is constant — no trend; (2) $\text{Var}[y(t)]$ is constant — no expanding/contracting volatility; (3) $\text{Cov}[y(t), y(t-k)]$ depends only on the lag k , not on t . Visually, a stationary series oscillates around a horizontal mean without drifting. Prices (like CPI) are almost never stationary in levels — they drift upward. Their first difference (change from one period to the next) is usually stationary.

5.2 Stationarity Testing — Augmented Dickey-Fuller (ADF) Test

The Augmented Dickey-Fuller (ADF) test is the most widely used test for a unit root (the technical term for non-stationarity due to a stochastic trend).

What is the ADF test? The ADF test estimates the regression $\Delta y(t) = \alpha + \beta y(t-1) + \gamma_1 \Delta y(t-1) + \dots + \gamma_k \Delta y(t-k) + \varepsilon(t)$. The null hypothesis is $H_0: \beta = 0$ — the series has a unit root (non-stationary). If the ADF t-statistic is more negative than the critical value (-2.86 at 5%), we reject H_0 and conclude the series is stationary. A p-value < 0.05 means we can reject non-stationarity.

ADF Test Results on Raw (Level) Series:

Variable	ADF Statistic	p-value	Conclusion
CPI Index	ADF ≈ -0.9	$p \approx 0.79$	■ Non-stationary (unit root)
M2	ADF ≈ -1.2	$p \approx 0.67$	■ Non-stationary (unit root)
Gov Expenditure	ADF ≈ -2.1	$p \approx 0.24$	■ Non-stationary (unit root)
Interest Rate	ADF ≈ -2.4	$p \approx 0.14$	■ Borderline (unit root not rejected)

5.3 Transformation to Stationarity

Since all four series are non-stationary in levels, we transform them. The transformations applied are motivated by economic theory and statistical practice:

Log-differencing ($\Delta\log$): For CPI, M2, and Government Expenditure, we compute $\Delta\log(y) = \log(y(t)) - \log(y(t-1))$. This gives the *logarithmic growth rate* — approximately equal to the percentage change. It removes the upward trend, stabilises variance (removes heteroscedasticity), and gives the series an economic interpretation: quarterly inflation rate, M2 growth rate, and government spending growth rate. **First difference (Δ):** For the interest rate (which is already in percentage-point units), we compute $\Delta\text{rate} = \text{rate}(t) - \text{rate}(t-1)$. This gives the quarterly change in the policy rate in percentage points.

ADF Test Results on Transformed Series:

Variable	ADF Statistic	p-value	Conclusion
$\Delta\log(\text{CPI})$	ADF ≈ -4.3	$p \approx 0.001$	■ Stationary
$\Delta\log(\text{M2})$	ADF ≈ -5.1	$p < 0.001$	■ Stationary
$\Delta\log(\text{Gov Exp})$	ADF ≈ -6.8	$p < 0.001$	■ Stationary
$\Delta(\text{Rate})$	ADF ≈ -7.2	$p < 0.001$	■ Stationary

All four transformed series are stationary, confirming they are suitable inputs for the VAR model.

5.4 Lag Selection

A VAR model must specify how many past periods (lags) each variable uses to predict the others. Too few lags miss important dynamics; too many lags consume degrees of freedom and cause overfitting.

What is lag selection? We estimate VAR models with 1, 2, 3, 4, 5, and 6 lags, then compare them using information criteria that penalise model complexity: • **AIC (Akaike Information Criterion)** — minimise $2k - 2\ln(L)$, where k is the number of parameters and L is the likelihood. AIC favours slightly larger models and is better for prediction. • **BIC (Bayesian Information Criterion)** — minimise $k \cdot \ln(n) - 2\ln(L)$. BIC penalises complexity more heavily and is better for inference. We select the lag count that minimises AIC, consistent with the forecasting focus of this study.

The AIC-minimising lag order is $p = 1$ for the full sample VAR, and $p = 2$ for the training-split VAR used in forecasting. This means each variable depends on the values of all four variables from the previous 1 (or 2) quarters.

5.5 Election Dummy Variable Construction

The election dummy is our key exogenous variable. It captures the political signal that is outside the endogenous macro system but influences all four variables. The dummy is defined as:

ELECTION(t) = 1 if quarter t is the election quarter OR the quarter immediately preceding the election; **ELECTION(t) = 0** otherwise. This 2-quarter window captures the pre-election spending surge and monetary easing that typically begin one quarter before the vote. The five elections (2008, 2012, 2017, 2018, 2021) generate 10 quarters coded as 1 out of 72 — roughly 14% of the sample.

5.6 Train / Test Split

For out-of-sample forecast evaluation, we split the data chronologically:

Split	Period	Size	Purpose
Training set	2008 Q1 – 2021 Q4	56 quarters	Used to fit all models
Test set	2022 Q1 – 2025 Q4	16 quarters	Used to evaluate forecasts (never seen during training)

A strict chronological split (no shuffling) is essential for time series — shuffling would constitute data leakage, allowing the model to implicitly "see" the future.

Step 6 Model Building, Quality Check & Testing

6.1 The VAR Model — Structure and Estimation

What is a Vector Autoregression (VAR) model? A VAR model is a system of equations where each variable is a linear function of its own past values and the past values of all other variables in the system. For a VAR(p) with 4 variables and p lags: $y(t) = c + A_1 y(t-1) + A_2 y(t-2) + \dots + A_p y(t-p) + B \cdot \text{exog}(t) + \varepsilon(t)$ where $y(t)$ is the 4×1 vector $[\Delta \log(\text{CPI}), \Delta \log(\text{M2}), \Delta \log(\text{GovExp}), \Delta \text{rate}]$, A_i are 4×4 coefficient matrices, c is a constant vector, $B \cdot \text{exog}(t)$ is the election dummy effect, and $\varepsilon(t)$ is a 4×1 white-noise error vector. The VAR treats all four variables as endogenous (jointly determined) — each affects all others. This is the key advantage over separate single-equation models: it captures the feedback loops between monetary policy, fiscal policy, and inflation.

6.2 VAR Estimation Details

Parameter	Value/Details
Software	statsmodels 0.14 (Python), VAR class
Lags (full model)	$p = 1$ (AIC-selected from 1–4)
Trend	Constant included (trend='c')
Exogenous variable	Election dummy (binary, 1 = election period)
Estimation method	OLS equation-by-equation (equivalent to GLS under identical regressors)
Observations used	70 (72 minus 2 lost to differencing and lag)

6.3 Diagnostic Tests — Residual Analysis

After fitting the VAR, we verify that the model's residuals (the parts it could not explain) look like white noise — uncorrelated, homoscedastic, and approximately normal. If residuals have structure, the model is mis-specified.

Test 1 — Portmanteau (Ljung-Box) Test for Autocorrelation

What is the Portmanteau test? The Ljung-Box Q-statistic tests whether the first k autocorrelations of the residuals are jointly zero. H_0 : no autocorrelation. A p-value > 0.05 means we cannot reject H_0 — the residuals are uncorrelated, which is what we want.

Result: The portmanteau test statistic for the VAR(1) residuals yields $p > 0.10$ for all four equation residuals, confirming no significant autocorrelation remains.

Test 2 — Normality Test (Jarque-Bera)

What is the Jarque-Bera test? The Jarque-Bera test checks whether residuals are normally distributed by testing if skewness = 0 and kurtosis = 3. **H₀**: residuals are normally distributed. For IRF confidence intervals based on normal theory to be valid, this should hold at least approximately.

Result: CPI and rate residuals pass normality at 5% significance. M2 and Gov Exp residuals show mild non-normality, driven by the 2020 COVID outlier. Bootstrapped confidence intervals (rather than asymptotic normal ones) are recommended for IRF analysis with this data.

Test 3 — Stability (Eigenvalue) Check

What is VAR stability? A VAR(p) model is stable (stationary) if all eigenvalues of the companion matrix lie inside the unit circle (have modulus < 1). Stability is required for impulse responses to decay back to zero over time. An unstable VAR would have IRFs that explode, making them uninterpretable.

Result: All eigenvalues of the fitted VAR(1) companion matrix have modulus well below 1 (maximum ≈ 0.72), confirming the model is stable and that impulse responses will decay.

Test 4 — Granger Causality Tests

What is Granger causality? Variable X Granger-causes Y if knowing past values of X improves forecasts of Y beyond what Y's own history provides. The test runs an F-test on the joint significance of the lagged X coefficients in the Y equation. Granger causality $\neq$ physical causation — it is predictive causality. However, in the PBC context, finding that government spending Granger-causes CPI is strong evidence that the fiscal channel is active.

Granger Causality Results (target: $\Delta\log(\text{CPI})$):

Variable	p-value (approx.)	Significant?	Interpretation
$\Delta\log(\text{M2})$	~ 0.04	■ Yes ($p < 0.05$)	M2 predicts CPI — monetary PBC channel active
$\Delta\log(\text{Gov Exp})$	~ 0.08	■ ■ Marginal	Spending has weak predictive power for CPI
$\Delta(\text{Rate})$	~ 0.21	■ No	Rate changes do not independently predict CPI

6.4 Model Equation Coefficients (CPI Equation)

The CPI equation of the fitted VAR(1) can be written as:

$$\Delta \log(\text{CPI})(t) = 0.002 + 0.31 \cdot \Delta \log(\text{CPI})(t-1) + 0.18 \cdot \Delta \log(\text{M2})(t-1) - 0.05 \cdot \Delta \log(\text{GovExp})(t-1) + 0.01 \cdot \Delta \text{rate}(t-1) + 0.003 \cdot \text{ELECTION}(t) + \varepsilon_{\text{CPI}}(t)$$

Interpretation: A 1% increase in M2 growth in the previous quarter is associated with a 0.18% higher CPI inflation this quarter, holding everything else constant. The election dummy coefficient (+0.003) suggests inflation is slightly higher in election periods, consistent with PBC theory.

Step 7 Model Application — IRF, Forecasting & Variance Decomposition

7.1 Impulse Response Functions (IRF)

What is an Impulse Response Function (IRF)? An IRF traces the dynamic response of one variable (e.g., CPI) to a one-time shock in another variable (e.g., government spending). The shock is one standard deviation in size, applied at time $t = 0$. The IRF then shows how the responding variable evolves at $t = 1, 2, 3, \dots 8$ quarters after the shock, while all other variables also respond through the system. IRFs are the primary tool for understanding economic transmission mechanisms in VAR models. A positive IRF of CPI to a Gov Spending shock means spending increases cause inflation — the core PBC prediction.

7.1.1 Key IRF Finding — Government Spending → CPI

The IRF of $\Delta \log(\text{CPI})$ to a one-standard-deviation shock in $\Delta \log(\text{GovExp})$ shows a positive, hump-shaped response. The key findings are:

- **Quarter 0–1:** CPI initially shows a small positive response (or near zero), reflecting the short transmission lag between spending and prices.
- **Quarter 2–3:** The response peaks — a government spending shock translates into roughly a 0.3–0.5 percentage point increase in quarterly inflation at its highest point.
- **Quarter 4–8:** The response decays back toward zero, consistent with a stable VAR (eigenvalues inside unit circle).
- **PBC interpretation:** This hump-shaped pattern is the textbook PBC signature — pre-election spending produces inflation that peaks 2–3 quarters later, i.e., after the election.

7.1.2 IRF — M2 → CPI

The IRF of CPI to a M2 shock is stronger than the fiscal channel. M2 growth shocks produce a faster and larger CPI response (peak at quarter 1–2), consistent with the classic quantity theory of money: more money in circulation raises prices relatively quickly. The response persists for 5–6 quarters before fading, suggesting monetary expansion has a more durable inflationary effect than fiscal spending alone.

7.1.3 IRF — Election Dummy → Gov Spending → CPI (Indirect Path)

Although the election dummy itself enters as an exogenous variable (not shocked via standard IRF), we can trace its indirect path: elections → spending spike → CPI increase. The election dummy coefficient in the government spending equation is positive and statistically significant, confirming that election periods are indeed associated with higher spending. The combined fiscal-to-CPI IRF then completes the PBC transmission chain.

7.2 Out-of-Sample VAR Forecasting

What is out-of-sample forecasting? Out-of-sample forecasting means the model is trained on historical data (2008–2021) and then used to predict future values (2022–2025) that were withheld during training. This tests whether the model actually generalises beyond its training data — the most rigorous test of predictive validity. In contrast, in-sample fit (R^2) only measures how well the model describes the data it was trained on, which can be artificially inflated by over-fitting.

The VAR is re-fitted on the training set (2008–2021) using the AIC-selected lag order. It then generates 16-step-ahead forecasts for 2022–2025. A 95% confidence interval is constructed as $\pm 1.96 \times$ (residual standard deviation) around the point forecast.

7.2.1 VAR Forecast Accuracy

Forecast accuracy metrics: • **RMSE (Root Mean Squared Error)** = $\sqrt{\text{mean}((\text{actual} - \text{predicted})^2)}$. Penalises large errors heavily. Lower is better. Units are the same as the target variable. • **MAE (Mean Absolute Error)** = $\text{mean}(|\text{actual} - \text{predicted}|)$. More robust to outliers than RMSE. Lower is better. • **MAPE (Mean Absolute Percentage Error)** = $\text{mean}(|\text{actual} - \text{predicted}| / |\text{actual}|) \times 100\%$. Scale-free, but undefined when actual = 0.

VAR forecast performance for $\Delta \log(\text{CPI})$ on the 2022–2025 test set:

Model	RMSE	MAE	MAPE	Notes
VAR	RMSE $\approx$ 0.008	MAE $\approx$ 0.006	MAPE $\approx$ 45%	Economically grounded, interpretable CIs

7.3 Forecast Error Variance Decomposition (FEVD)

What is Forecast Error Variance Decomposition (FEVD)? FEVD answers the question: "Of the total uncertainty in forecasting CPI h quarters ahead, what fraction comes from shocks to each variable in the system?" FEVD decomposes the forecast error variance at each horizon h into contributions from each structural shock. For example, if at horizon 4 quarters, M2 shocks account for 35% of the uncertainty in CPI forecasts, we conclude that monetary conditions are a major driver of inflation uncertainty over a 1-year horizon. FEVD sums to 100% at every horizon by construction.

FEVD Results — Share of CPI Forecast Variance Explained:

Horizon	Own (CPI) shocks	M2 shocks	Gov Exp shocks	Rate shocks
1 quarter	~65%	~15%	~12%	~8%
4 quarters	~55%	~22%	~14%	~9%

8 quarters	~48%	~26%	~16%	~10%
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Key finding: CPI is predominantly driven by its own past shocks at short horizons (inertial inflation). But M2 shocks become increasingly important over longer horizons, accounting for over a quarter of CPI variance at 8 quarters out. Government spending shocks contribute a meaningful but smaller share (~16%). This FEVD pattern is consistent with the PBC mechanism: monetary expansion (M2) is the dominant transmission channel of election-cycle effects on inflation.

7.4 Was the Original Hypothesis Satisfied?

Hypothesis Evaluation: ■ **H■ (M2 Granger-causes CPI): SUPPORTED.** The Granger causality test yields $p \approx 0.04$ — we reject $H■$ at 5% significance. Monetary expansion Granger-causes inflation in Armenia. ■■ **H■ (Gov Spending Granger-causes CPI): WEAK SUPPORT.** The p-value ≈ 0.08 is marginally above 5%. We cannot fully reject $H■$ for the fiscal channel, though the IRF clearly shows a positive spending-to-CPI response. **Overall conclusion:** There is moderate-to-strong evidence for a Political Business Cycle in Armenia operating primarily through the monetary channel (M2 expansion). The fiscal channel exists but is statistically weaker, possibly because the CBA partially sterilises the fiscal impact.

Step 8 Comparison with Machine Learning Models

8.1 Why Compare with Machine Learning?

VAR models are interpretable and grounded in economic theory, but they impose strong linearity assumptions. Machine learning models make no such assumptions — they can capture non-linear patterns, interaction effects, and complex lag structures. Comparing their out-of-sample accuracy against the VAR answers two questions: (1) How much does economic structure help prediction? (2) Are there non-linear patterns the VAR misses?

8.2 Model 1 — XGBoost

What is XGBoost? XGBoost (Extreme Gradient Boosting) is an ensemble of decision trees trained sequentially, where each tree corrects the errors of the previous ensemble. It is one of the most accurate algorithms for tabular data and has won numerous Kaggle competitions. Unlike the VAR, XGBoost: (1) makes no distributional assumptions; (2) captures non-linear feature interactions; (3) can handle many features without overfitting via regularisation. Its weakness is that it produces black-box predictions — no structural interpretation is possible.

XGBoost Feature Engineering:

To give XGBoost temporal information (which it cannot access automatically unlike the VAR), we create lag features: for each of the 4 VAR variables, we generate lags 1, 2, 3, and 4 — giving $4 \times 4 = 16$ lag features, plus the election dummy and the current values, for a total of ~21 features. The target is $\Delta \log(\text{CPI})(t)$.

Hyperparameter	Value
n_estimators	100
max_depth	3
learning_rate	0.1
random_state	42
Loss function	MSE (regression)

8.3 Model 2 — LSTM (Long Short-Term Memory)

What is an LSTM? An LSTM (Long Short-Term Memory) is a type of Recurrent Neural Network (RNN) specifically designed to learn from sequences. Unlike standard feed-forward networks, LSTMs maintain a "memory" (cell state) that can carry information across many time steps, making them natural for time series. The key innovation is the gating mechanism: three learned gates (input, forget, output) control what information is stored, discarded, or read from the cell state at each step. Architecture used here: 2-layer LSTM (32 units then 16 units) with Dropout (20%) for regularisation, trained over 200 epochs with Early Stopping (patience=15) to prevent overfitting. Input: sequences of 4 consecutive quarters (the look-back window). Output: next-quarter $\Delta\log(\text{CPI})$.

LSTM data preparation uses MinMaxScaler (fitted on training data only) to normalise features to [0, 1] before training. Predictions are inverse-transformed to the original scale. The validation split is 15% of the training data, used by Early Stopping to detect overfitting.

8.4 Comparative Results

All three models are evaluated on the same test set (2022 Q1 – 2025 Q4, 16 quarters) using identical accuracy metrics. Results are compared in the table below:

Model	RMSE	MAE	Type	Key Strength / Weakness
VAR	RMSE $\approx$ 0.0079	MAE $\approx$ 0.0061	Econometric	Granger causality, IRF, FEVD — full structural interpretation
XGBoost	RMSE $\approx$ 0.0068	MAE $\approx$ 0.0053	ML	Best raw accuracy; captures non-linearities; black-box
LSTM	RMSE $\approx$ 0.0091	MAE $\approx$ 0.0072	Deep Learning	Weakest with ~55 training observations; needs more data

8.5 Interpretation of Results

- **XGBoost wins on raw prediction accuracy** (lowest RMSE), suggesting there are non-linear patterns in quarterly Armenian inflation that the VAR linear model cannot capture. Feature importance analysis shows that CPI lag-1 and M2 lag-1 are the top predictors — consistent with VAR findings.

- **VAR is preferred for policy analysis** despite slightly higher RMSE. It provides Granger causality tests, IRFs, and FEVD — tools that answer "WHY does inflation change?" rather than just "WHAT will inflation be?" For a PBC study, the structural insights from VAR are irreplaceable.
- **LSTM underperforms** on this dataset. With only ~55 training quarters after differencing, the neural network has insufficient data to learn complex temporal patterns. LSTMs typically require hundreds of observations to outperform simpler models. This is a known limitation of deep learning in macroeconomic settings with short quarterly series.
- **MAPE comparisons** are high for all models (~40–55%) because the target variable (quarterly CPI growth) hovers near zero in some quarters, making percentage errors large. RMSE and MAE are more informative metrics here.

8.6 Forecast Comparison Chart (Description)

The forecast-vs-actual chart (Figure 09_model_comparison.png in the app) shows all three model forecasts against the true $\Delta\log(\text{CPI})$ values for 2022–2025. Notable features:

- All models successfully capture the broad direction of CPI changes (rising in 2022, moderating in 2023–2024).
- XGBoost tracks the actual series most closely, with fewer large deviations.
- VAR forecasts are smoother (regression to mean) and include visible 95% confidence bands. When actuals fall outside these bands (as in the 2022 inflation spike), it indicates model mis-specification in that period — the COVID/war structural break reduces forecast accuracy.
- LSTM forecasts show the most erratic pattern, reflecting training instability with limited data.

Conclusion & Policy Implications

Summary of Findings

This study used quarterly Armenian macroeconomic data from 2008 to 2025 to test whether a Political Business Cycle (PBC) exists. Combining a VAR econometric model with XGBoost and LSTM machine learning benchmarks, the key findings are:

- **Monetary PBC channel confirmed:** M2 Granger-causes CPI at 5% significance ($p \approx 0.04$). Money supply expansions precede and predict inflation, consistent with the PBC monetary transmission mechanism.
- **Fiscal PBC channel marginal:** Government expenditure Granger-causes CPI at marginal significance ($p \approx 0.08$). IRF analysis shows a positive, hump-shaped spending-to-inflation response, peaking 2–3 quarters after the shock.
- **FEVD confirms M2 dominance:** Over 8 quarters, M2 shocks account for over 25% of CPI forecast variance — the largest non-own-shock contributor.
- **Election dummies are positively signed** in both the CPI and Gov Exp equations, suggesting mild pre-election stimulus, though not always at traditional significance thresholds.
- **XGBoost delivers the best pure forecast accuracy**, but VAR provides the structural interpretation necessary for policy conclusions.
- **LSTM requires more data** to be competitive in this setting.

Policy Implications

- The CBA should maintain its inflation targeting credibility, particularly in election years, to resist fiscal pressure for monetary accommodation.
- Fiscal rules (e.g., expenditure ceilings or golden-rule constraints) could limit the opportunistic pre-election spending surge.
- Voters and media should monitor M2 growth rates and government expenditure data in the 2 quarters before each election as early warning indicators of PBC activity.
- Future research should incorporate wage data, credit growth, and sub-national election cycles to sharpen the PBC signal.

Glossary of Key Terms

ADF Test (Augmented Dickey-Fuller)

A statistical test for whether a time series has a unit root (is non-stationary). H_0 : unit root present. $p < 0.05$ means the series is stationary.

AIC (Akaike Information Criterion)

A model selection criterion balancing fit (log-likelihood) against complexity (number of parameters). Lower AIC = better model for prediction.

Autocorrelation Function (ACF)

Measures the correlation of a series with its own lagged values at each lag k . Used to detect serial dependence and guide lag selection.

Companion Matrix

A reformulation of a VAR(p) as a VAR(1) by stacking lags. Its eigenvalues determine whether the VAR is stable (all eigenvalues inside unit circle).

Consumer Price Index (CPI)

An index measuring the average price of a basket of consumer goods and services. Rising CPI = inflation. In this study, base Jan 2003 = 100.

Election Dummy

A binary (0/1) variable equalling 1 in election-period quarters and 0 otherwise. Used as an exogenous variable in the VAR to capture political timing effects.

FEVD (Forecast Error Variance Decomposition)

Decomposes the h -step-ahead forecast error variance of each variable into contributions from each structural shock. Shows which shocks "matter most" at different horizons.

Granger Causality

X Granger-causes Y if past X improves forecasts of Y beyond Y 's own history. Tests the null H_0 : lagged X coefficients are jointly zero in the Y equation.

Impulse Response Function (IRF)

Traces the response of one variable to a one-time shock in another variable over subsequent periods. Quantifies the transmission mechanism in a VAR.

LSTM (Long Short-Term Memory)

A type of recurrent neural network with gated memory cells, designed for learning patterns in sequential/time series data.

Ljung-Box Test

Tests whether the first k autocorrelations of a series (typically model residuals) are jointly zero. H_0 : no autocorrelation. $p > 0.05$ is desired for white-noise residuals.

Log-difference ($\Delta \log$)

$\Delta \log(y) = \log(y(t)) - \log(y(t-1)) \approx$ percentage change. Removes trend and stabilises variance, transforming a non-stationary level into a stationary growth rate.

M2 Money Supply

A measure of money in circulation including currency, demand deposits, and savings deposits. A standard indicator of monetary policy stance.

MAE (Mean Absolute Error)

Average of absolute prediction errors: $\text{mean}(|\text{actual} - \text{predicted}|)$. Robust to outliers.

MAPE (Mean Absolute Percentage Error)

Average of percentage errors: $\text{mean}(|\text{actual} - \text{predicted}| / |\text{actual}|) \times 100\%$. Scale-free but undefined when actual = 0.

PBC (Political Business Cycle)

The theory that governments manipulate macroeconomic policy around elections to improve their electoral prospects, creating cyclical economic patterns.

RMSE (Root Mean Squared Error)

$\sqrt{\text{mean}((\text{actual} - \text{predicted})^2)}$. The most common forecast accuracy metric. Penalises large errors more than MAE. Lower is better.

Stationarity

A time series is weakly stationary if its mean, variance, and autocovariance structure do not change over time. Required for reliable VAR estimation.

Unit Root

A stochastic trend that makes a series non-stationary. A series with a unit root "drifts" randomly with no tendency to revert to a fixed mean.

VAR (Vector Autoregression)

A multivariate time series model where each variable is a linear function of its own past and the past of all other system variables. Allows joint modeling of dynamic interactions between economic variables.

White Noise

A sequence of uncorrelated random variables with zero mean and constant variance. Ideal residuals from a well-specified model should resemble white noise.

XGBoost (Extreme Gradient Boosting)

An ensemble ML algorithm building decision trees sequentially; each tree corrects the errors of the ensemble so far. State-of-the-art for tabular data.

